TABLE I
OPTICAL PROPERTIES AND BODY METABOLITES

Absorbance by skin due to chemical components	Transmittance wavelength is a function of	BG measurement determined by optical signal variations in	Other factors in BG measurement {site (%) of fluid sample}
Water	Thickness	Wavelength	Interstitial
Haemoglobin	Colour	Polarisation	Intracellular
Melanin	Structure of skin	Intensity of light	Capillary blood
Fat	Bone	-	Ratio of tissue fluid
Glucose	Blood	-	Activity level
-	Other materials	-	Diet or hormone variations

indirect blood contact meters. Non-invasive blood glucose measurements are like the fairy tale of Rapunzel—a princess who acquires light emanating and non-invasive healing powers in her orbicular long hair.

Non-invasive BG meter technology based on the optoelectronics principle is free of blood samples and, thus, pain-free. Ideally, tissue samples are taken from the sides of fingers, fingertips or earlobes for non-invasive BG measurement.

Optical non-invasive system for BG measurements. The technology used for glucose measurement is based on the principle that the human body naturally emits strong electromagnetic radiation in the micrometre wavelength region, and the discovery that radiation contains spectral information of tissue analytes.

The laws of Physics state that all objects emit infrared (IR) radiation and that the intensity of the radiation and spectral characteristics of the object are determined by its absolute temperature as well as by the properties and states of the object.

Non-invasive BG measurement system. The non-invasive BG measurement system, based on the principle of optoelectronics, typically consists of: non-invasive (sensor-transducer optoelectronics system) and optical sources (like photo detectors, and signal conditioning and acquisition systems).

Non-invasive optical measurement of glucose is performed *in vivo* by focusing a beam of light on the body. The light is modified by the tissue after transmission through the target area. An optical signature or fingerprint of the tissue content is produced by the diffused light that shuns the penetrated tissue. Details given in Table I help in analysing the optical response of body metabolites in terms of their absorbance and transmittance.

Evaluation of a BG value of 100mg/dl is equivalent to a tissue sample glucose average of 38mg/dl of which 26 per cent is due to blood, 58 per cent due to interstitial fluid and 16 per cent due to intra-cellular fluid. Non-invasive optical measurement sometimes includes light sources or non-optical sources such as sweat, urine or tears samples for analysis *in vitro*.

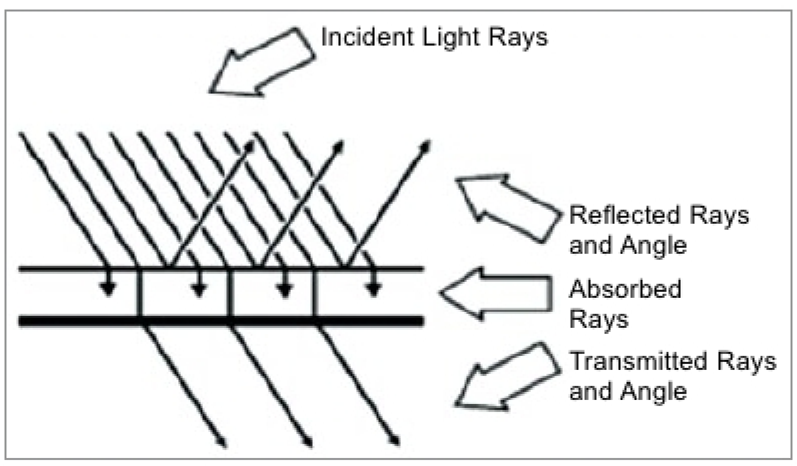

Fig. 2: Optical properties

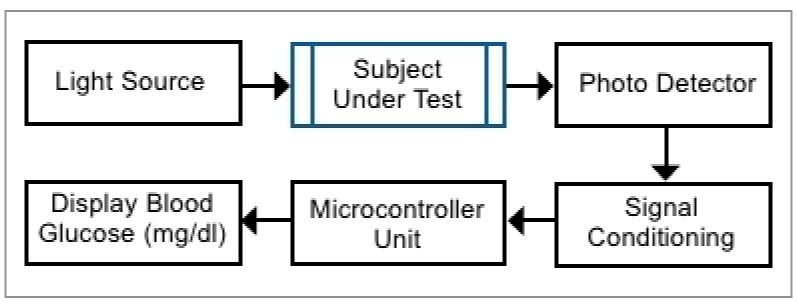

Fig. 3: Block schematic of optical measurement system